

Questions:

- Q1) Notebook downloads imagery live from server instead of a satellite image file as they are massive and highly impractical to share or ship alongside a code notebook.
- Q2) a) The number of tiles downloaded would increase.
b) The model will see small rooftops much more clearly because the resolution improves from roughly 0.6m/pixel to 0.3m/pixel .
- Q3) $\rightarrow$ Lowering to 0.4 : The building count will increase, risking false positives (counting fake buildings/noise)
 $\rightarrow$ Raising to 0.9 : The building count will decrease, risking false negatives (missing real buildings)
- Q4) Fine-tuning allows the model to leverage pre-existing low-level feature detectors (like edges, shapes and textures) already learned in sandbox rather than learning them from scratch.
- Q6) The mAP (mean Average Precision) metric relies heavily on a perfect "answer key" (ground truth labels). If the label dataset itself has missing buildings or slight mismatches, the model gets penalized mathematically even if its visual predictions are highly accurate.
- Q7) A simple building count completely misses the vertical density (no. of floors) and the size of buildings.

Q8) Building the model was not a waste of time because it provided critical foundational knowledge on managing the "domain gap" and executing transfer learning. Understanding how to customize weights through fine-tuning is highly required when no pre-existing dataset is available. While Google Open Buildings serves as a superior dataset for this task, relying only on existing tools without knowing how to train serves no purpose.

Code - Questions :

Q1) The sandbox model suffered from a severe "domain gap" because it was trained entirely on SpaceNet data featuring large, regular, well-separated western buildings. When deployed on Ucknow, it encountered small, irregular dense Indian rooftops that it had never seen before, causing it to fail.

Q2) The model roughly detected 4638 buildings. It missed the most buildings in the densest slums and highly congested neighborhoods where individual rooftops touch wall-to-wall and become visually indistinguishable.

Q3) Open Buildings performed significantly better because it is the finished output of a much larger, more powerful AI model run at a scale by Google across high resolution imagery.